

information such as the e-mail address from where the mail came, the size of the mail, attachments, and the time it was sent and received by you.

The Details tab is where you'll see the e-mail header. This is what a typical e-mail header looks like.

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Return-Path: <achanen@bemybabey.com>
Delivered-To: mohank@mmb4.vsnl.net.in
Received: from isa.es.kr (unknown
[211.184.249.61]) by mmb4.vsnl.net.in
(Postfix) with SMTP id 34621EE76 for
<mohank@giasbm01.vsnl.net.in>; Wed, 12
Dec 2001 05:43:22 +0530 (IST)
Received: from bronxcra.cx (unknown
[233.144.259.67]) by isa.es.kr (Postfix) with
SMTP id 34644BB76 for <mohank@gias-
bm01.vsnl.net.in>; Wed, 12 Dec 2001
05:43:22 +0530 (IST)
Received: (qmail 6042 invoked from net-
work); 27 Aug 2001 11:17:09 -0000
Received: from pppa96-
resalechicagometro13-5r7120.dialinx.net
(HELO gw08??192.168.227.32?)
(4.4.242.29) by 211.184.249.61 with
SMTP; 27 Aug 2001 11:17:09 -0000
Received: from mail4.bemybabey.com by
gw08 with ESMTP; Mon, 27 Aug 2001
06:10:54 -0400
Message-ID:
<00007a84395a$00001903$00005320@mail4.bemybabey.com>
To: <achanen@bemybabey.com>
From: achanen@bemybabey.com
Subject: Make It Happen! (Email Special)
21280
Date: Mon, 27 Aug 2001 06:10:52 -0400
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The Received: header entry contains the site from which the mail was sent and the IP address from where the mail originated. The IP address could be of the computer that's connected to the Net or more often the IP address of an ISP or the proxy server (that is if the e-mail was sent from a corporate LAN). It also contains information about the SMTP servers or other Web sites through which the mail passed on its way to your computer.

The important thing to realise here is that if a person is faking his e-mail address or the SMTP servers through which it passes, then there would be

Sniffing out the Spammer

In a cyber crime, it's the computer and the storage media within the computer, such as the hard disk, that hold the key to uncovering the mystery. Cyber-forensics or computer forensic science is a process of extracting data from computer storage media in an organised manner using a set of tools and techniques. It's certainly not a plug-and-play operation or a collection of just physical evidence or data. It also involves sifting through enormous amounts of technological data collected; interviewing suspects, narrowing down suspects and piecing together a trail of evidence that holds good in a court of law.

Forensics analyses involves examining free space on hard disks, examining the file slack (the space from the end of a file to the end of a storage cluster) for hidden

or deleted data, analysing Windows swap files, etc. In fact it's possible for cyber-forensics experts to even locate files that have been consciously deleted.

The various tools and forensics software used in a cyber-forensics analysis work with specific platforms such as Unix, Linux, Windows, Macintosh, etc. They include software such as DiskScrub, DiskSearch or Net Threat Analyser from companies like New Technologies Inc. (www.forensics-intl.com) or enCase from Guidance Software (www.guidance-software.com).

In India, the Mumbai Cyber Crime Cell works with Internet security consultants such as CyberForensics India to help uncover cyber crimes like hacking and online credit card fraud.

many additional Received entries. In the header above, most likely the mail has been sent from Isa.es.kr and the rest of them are false headers. Using the **tracert** command, you can check which headers are fake and which are genuine.

The headers also contain information about the time (mostly local time) and the date the e-mail passed through that particular server. The header has a message ID, which is a unique ID assigned to a message. The message ID here is: 00007a84395a\$00001903\$00005320@mail4.bemybabey.com

The latter half of the message ID would contain the Web site from where the mail came or an IP address.

What if I use Web mail?

You can still check the e-mail header. Most e-mail service providers such as Yahoo! or Hotmail, by default, don't display the entire message header, but they have an option to enable display of message headers. In Hotmail, click on Options and check Mail Display Settings. You can choose to not display it at all or choose from Basic, Full or Advanced header display options.

At Yahoo! you'll find message header options at Options under Mail Preferences. You have a choice between Brief or Full header display.

Unfortunately, not all mail services let you view the header information when you need to complain to the site from where the e-mail was sent or report the matter to the police.

Performing a Traceroute

The Received: entry could either contain the name of a Web site or an IP address. To find out who the IP address belongs to or the name of the site, you could look up the IP address at a database of IP addresses. For example, a 'whois' search on the address at a site such as www.internic.net/whois.html will help you find out the registrar of the domain name or the e-mail address of the administrator.

You could also run a Traceroute on the IP address to trace the e-mail to the host Web site or server. Traceroute utilities track the path taken by packets as they travel from source to destination. A Traceroute can be run from Windows Start at **Accessories > Command Prompt**.